

1.7 Quiz: Measurement

Reference (provided). average speed = $\frac{\text{distance}}{\text{time}}$

1 m $\approx$ 3.28 ft 1 mi = 5280 ft 1 km = 1000 m 1 m = 100 cm 1 cm = 10 mm
1 h = 3600 s 1 min = 60 s 1 gal $\approx$ 3.785 L

1. Write down the number of significant digits in each measurement.

(a) 0.0350 m

(c) 7.0 s

(e) 0.000 09 kg

(b) 806 g

(d) 12.50 cm

(f) 6.02×10^{23}

2. Round each value to three significant figures. Copy the calculator display followed by three dots, then round: $\sqrt{2} = 1.4142135 \dots \approx 1.41$

(a) $\sqrt{13}$

(b) The height of Mount Everest,
8848.86 m

3. Write in scientific notation, rounded to three significant figures.

(a) The distance light travels in one year,
about 9,460,730,000,000 km

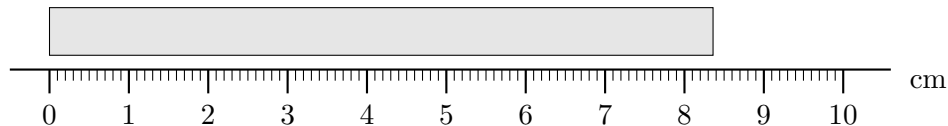
(b) The charge of an electron,
0.000 000 000 000 000 160 2 C

4. Write each value as an ordinary decimal number.

(a) The width of a human hair,
 7.0×10^{-5} m

(b) The closest distance from the Earth
to Mars, 5.46×10^7 km

5. The bar below is measured with a centimeter ruler marked in millimeters.



- (a) Record the length of the bar. Include one estimated digit — the last digit, the one you are not certain of.
- (b) How many significant figures does your measurement have?
- (c) Write your measurement in millimeters.
6. Convert between units, rounding to three significant figures. Write each conversion as a single chain of fractions and show the units cancelling.
- (a) Taipei 101 is 508 meters tall. What is its height in feet?
- (b) Spain's AVE high-speed train runs at 310 kilometers per hour. What is this speed in meters per second?
7. A cheetah has been timed running at 29 meters per second. What is this speed in miles per hour?

8. Mars is 5.46×10^7 km from the Earth at its closest. A radio command sent to a rover on Mars travels at the speed of light, 3.00×10^8 m/s. How many minutes does the command take to arrive?
9. The Atlantic Ocean widens by about 2.5 centimeters per year. Columbus first crossed it in 1492. By how many meters has the Atlantic widened between 1492 and 2026?

10. A US shower head lets through at most 2.5 gallons of water per minute. How many liters of water does an 8.0-minute shower use?
11. New York City uses about 1.0 billion gallons of water per day, for a population of about 8.5 million people. How many liters of water is that per person per day? Round your answer to the number of significant figures the data supports.